



Research paper

Evaluation of effect of providing balancing reserves to power system in Japan through operational control of water pumps

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ABSTRACT

In Japan, renewable energy is being rapidly introduced. Hence, surplus power and the required balancing reserves will increase. In this study, to deal with this problem, we focus on a water distribution system (WDS), especially water pumps, which can contribute to supply–demand balancing by shifting their operation. To evaluate the current potential of water pumps in Japan, we modeled them and simulated their operation, in which they can shift their pumping schedule to reduce surplus power and provide balancing reserves to the power system, adhering to constraints such as the demand for water and the storage of water reservoirs. We then developed a numerical model that simultaneously optimizes the operation of generators and water pumps throughout Japan, and simulated their operation in 2040. The results showed that despite increasing their power consumption, pumps shifted their pumping schedule to contribute to the entire power system and reduced fossil generator fuel costs and renewable energy output curtailment.

1. Introduction

The introduction of variable renewable energy (VRE) power generation, such as photovoltaic (PV) and wind power (WP), is progressing rapidly in Japan. This phenomenon was triggered by the introduction of the Excess Electricity Purchasing Scheme for PVs launched in 2009 and the Feed-in Tariff program launched in 2012, under which electric utilities purchase electricity generated by VRE power generation at a set price. As of the end of FY2021, the amount of PV and WP installed was 69.35 GW and 4.58 GW, respectively (Agency for Natural Resources and Energy, JAPAN, 2023). Compared to other countries, Japan has a much higher rate of introducing PV systems than wind turbines. Therefore, to make most effective use of VRE, it is necessary to utilize daytime electricity generated by PV on sunny days, which exceeds normal electricity demand.

In addition, the amount of electricity generated by VRE power generators depends on weather conditions and is thus characterized by unstable output. In the current power system, thermal power generation is mainly used to provide supply–demand adjustment power to cope with output fluctuations of VRE generation and fluctuations in electricity demand. The supply–demand adjustment capability in thermal power generators includes a governor-free operation, also called the primary control, in which the generator's regulator automatically

changes the output in response to frequency fluctuations with a cycle of up to a dozen seconds; a load frequency control (LFC), also called the secondary control, in which instructions are sent from the feeder control station to the power plants to control output in response to frequency demand fluctuations with a cycle of several minutes to several dozen minutes; and an economic load dispatching control, also called the tertiary control, in which the economic load is distributed to each power plant in response to frequency demand fluctuations with a cycle of several dozen minutes or longer. As the introduction of VRE generation increases, short-period fluctuations in VRE output increase, meanwhile supply–demand adjustment capabilities of thermal power generators decrease, because the number of operating thermal power generators is decrease. As a result, there are growing concerns about the shortage of supply–demand adjustment capability to short-term fluctuations, especially in the LFC domain. In order to expand the introduction of VRE, it is essential to secure LFC regulating reserves to replace thermal power generators.

In order to effectively utilize the daytime surplus electricity, and to secure additional LFC regulating reserves, demand response (DR) using distributed energy resources (DER) on demand side, such as stationary storage batteries, electric vehicles, residential heat pump water heaters, and other demand side equipment, is expected. Many

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studies have been conducted on the contribution of stationary storage batteries (Guo et al., 2021), electric vehicles (Iwafune et al., 2019), and heat pump water heaters (Omori et al., 2020) to power system operation through DR. In this study, we focused on the DR using water distribution systems (WDSs) as DER.

The WDSs in Japan require a large amount of electricity to operate water purification facilities and pump water to high elevations, and WDS projects account for approximately 1% of the nation's electricity consumption (Ministry of Health, Labour and Welfare, 2009). Water pumps are used in the process of transporting water from rivers to distribution reservoirs after converting it to purified water, and energy consumption in this process accounts for a large portion of the total energy consumption of WDSs. Many water utilities in Japan that consume large amounts of electricity have already introduced high-efficiency equipment, inverter control of pumps, and storage batteries for peak shaving, as energy conservation measures. In the future, effective use of VRE will be required for further environmental and energy conservation measures (Ministry of Health, Labour and Welfare, 2013).

In the process in which water pumps send water from the water treatment plant to the distribution reservoirs, the operating timing of the pumps can be changed because the distribution reservoirs are used as energy storage devices. Therefore, By shifting the operating hours of the pumps to daytime hours when PV generation is in surplus, it is expected that VRE can be used more effectively. Furthermore, water pumps can quickly change the amount of water delivered through inverter control. Since power consumption can be quickly increased or decreased, it is also possible to have a supply–demand adjustment capability in the LFC domain for the power system operation. From these points, WDS has high potential to contribute to power system operation by shifting the timing of electricity demand and providing LFC regulating reserves.

Several studies have been conducted on the operational control and modeling of the pumps in the WDS. As research into methods of saving energy and reducing energy costs for water supply and distribution systems, the optimization of determining the operation of water pumps to minimize their one-day power consumption by modeling them as mixed integer linear programming (MILP) was introduced (Arai et al., 2012). In addition, a pump operation planning model for a large-scale water distribution network (Hirano et al., 2017), minimizing the overall operating cost of a real WDS with the flexible operation of water pumps and in-house pumps at a water treatment plant, was proposed (Zimmermann et al., 2018). In addition, a model that considers the physical constraints of the entire system was proposed (Fooladivanda and Taylor, 2018). Various methods have also been proposed to contribute to electricity supply and demand by controlling the operation of water pumps in WDSs (Takahashi et al., 2016; Menke et al., 2016; Negishi and Ikegami, 2019; Mkireb et al., 2019; Negishi and Ikegami, 2021). For example, assuming a DR request to the WDS, an operation to minimize the peak power of the grid was modeled (Takahashi et al., 2016). In addition, assuming an incentive-type DR, a model was proposed to minimize the net cost of the DR by subtracting the benefit of the DR from the cost of electricity during pump operation (Menke et al., 2016). Additionally, a method was proposed for modeling and optimizing the operation of water pumps that provide frequency regulating power to the power system as MILP (Negishi and Ikegami, 2019). A control method was also proposed to ensure the stable operation of water pumps when there is uncertainty about water demand and the provision of frequency regulating power (Mkireb et al., 2019; Negishi and Ikegami, 2021). Furthermore, some studies have evaluated the impact of changing the operation of water pumps for WDSs on the system. In Refs. Imanaka and Uezu (2015) and Imanaka et al. (2016), the impact of changing the operation of water pumps in a closed power system on an actual island on energy conservation for the entire island was demonstrated and evaluated. Various research and optimization

models have been proposed to evaluate the efficiency of power consumption of pumps in WDSs. However, these studies are characterized by dealing with optimal operation for individual pumps but not for a whole power system, dealing with the contribution through peak cuts but not through the provision of short-term regulating reserves, or dealing with small-scale grids but not large-scale multiple power systems like Japan as a whole.

This study aimed to develop an optimization model that integrally handles the DR operations using all of Japan's WDSs and the operations of 10 power systems in Japan, and to quantitatively evaluate the economic and environmental contribution potential of the DR operations using these WDSs. In this study, two types of DR operations were considered: a shift in operating hours to utilize surplus daytime PV generation, and an operation in which pump inverter control provides short-term regulating reserved in the LFC domain. The first feature of this research is that the effects of these two types of DR operations of WDSs were evaluated using the Supply-Demand Analysis Model for optimizing generator operation across multiple power systems connected by wide-area interconnection lines.

Japan is part of the Pacific Ring of Fire and is geographically characterized by mountainous terrain, so there are large differences in elevation between regions. In addition, there is concentration and dispersion of the population depending on the region. The geographical characteristics of each region of Japan also affect the size of the pumps and distribution reservoirs of water treatment plants that supply water to that region. Therefore, it is not possible to evaluate the potential of the DR operations in Japan as a whole unless the pump systems and their operations in WDS in various regions of Japan are simulated. On the other hand, the number of WDSs in Japan far exceeds the number of the generators in all power systems. Therefore, handling all WDSs individually increases the scale of the model, and is difficult from the perspective of computational load.

In this study, we classified all WDSs in Japan into four clusters according to their geographical characteristics, and the systems and the operations of four WDSs that represent the clusters were incorporated into the power system Supply-Demand Analysis Model. Using this novel method, we succeeded in developing a model that can perform optimization within an executable time range. The second feature of this study is that a large number of WDSs throughout Japan were clustered based on geographical characteristics, and their operations were incorporated into the power supply–demand analysis model.

The remainder of this paper is organized as follows. The analytical methodology used in this study is described in Section 2. Section 3 describes the modeling of the WDSs used in the analysis for all of Japan. Section 4 describes the computational model used. Section 5 describes the computational conditions and results of the case analysis and discusses the potential impact of the Japanese WDS on the power system. Section 6 presents conclusions based on the contents of this study.

2. Analytical methods

2.1. Outline of analysis

The overall flow of analysis performed to evaluate the contribution of water pumps to the Japanese power system is shown in Fig. 1.

First, we obtained the elevation difference and population density data in each water supply area in Japan, which is based on the geographic information system (GIS). The water supply areas of water utilities throughout Japan are classified into four groups based on two parameters: the difference in elevation within the water supply area and the population density of the water supply population. We determined the water service providers that represent these four groups, and determined the water supply unit configurations for the four groups. We calculated the equivalent number of each of the four types of WDS

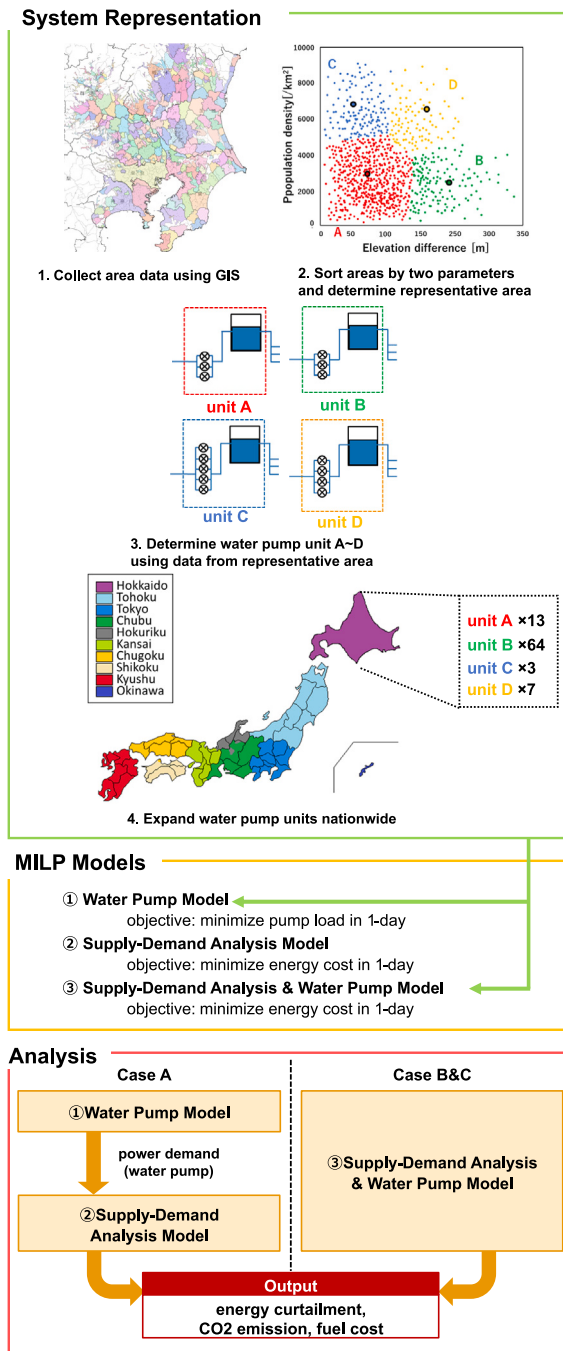


Fig. 1. Outline of analysis.

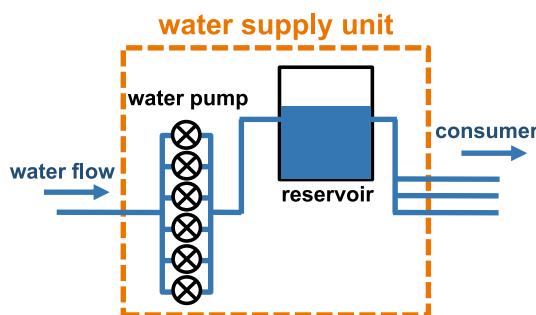


Fig. 2. Schematic diagram of water supply unit.

that exist within Japan's 10 power system regions. Details of these methods are described in Section 2.2 and Section 4.

Next, we developed the three models formulated as MILP: the Water Pump Model, the Supply-Demand Analysis Model, and the Supply-Demand Analysis with Water Pump Model. Details of these models are described in Section 2.3 and Section 4.

Assuming that the current water pump is operated under the minimization of power consumption, we calculated the power consumption of the current water pumps using the Water Pump Model, and then based on these power consumption, we performed the Supply-Demand Analysis Model, which can calculate all generators operation and inter-connection line power flow in Japan's 10 power systems, to calculate the amount of VRE output curtailment, CO₂ emissions, fuel costs, etc. This analysis was designated as Case A. This calculation result represents the base case before DR operation using the control of the pumps in WDSs.

Finally, using the Supply-Demand Analysis with Water Pump Model, which incorporates water pump operation into the power supply-demand analysis model, we performed calculations for DR operation using the control of the water pumps in WDSs. At this time, calculations were performed for Case B, which only uses DR operation by shifting pump operation hours, and Case C, which also uses DR operation that provides LFC regulating reserves through inverter control. By comparing these calculation results with current operation results of Case A, we examined the impacts and the potentials of DR operations using water pumps in all WDSs on the Japanese 10 power systems.

2.2. System representation

To simulate Japanese WDSs that considers the characteristics of the topography and population distribution, we classified the water supply areas of water suppliers nationwide into four groups from A to D using two parameters: the difference in elevation within the water supply area and the population density of the water supply population.

Firstly, as shown “System Representation” No. 1 in Fig. 1, we collected the GIS data, which has the elevation difference and population density of the water supply area of each water supplier nationwide from the Ministry of Land, Infrastructure, Transport, and Tourism's National Land Information (i.e., data on waterwork-related facilities, elevation, and slope 5th order mesh data) (Ministry of Land, Infrastructure, Transport and Tourism, 2012, 2011).

Secondly, as shown “System Representation” No. 2 in Fig. 1, the elevation difference and population density of the water supply area for each water utility were calculated. The distribution of water service providers based on the two parameters of elevation difference and population density of water supply was divided into four groups using the k-means++ method (Arthur and Vassilvitskii, 2007), a clustering method.

Next, as shown “System Representation” No. 3 in Fig. 1, the water service providers and their water supply areas closest to the last center of gravity of the clustering were designated as representative areas in the groups. In this paper, we assumed a single system in which multiple water pumps connected in parallel deliver water to a single distribution reservoir, and water is distributed from the reservoir according to the amount of water demand, as shown in Fig. 2. In this system, water from a water treatment plant is delivered to the distribution reservoir by water delivery pumps, and water delivery pumps consume electricity. In this paper, the water pump and distribution reservoir are collectively referred to as the “water supply unit”. From the water supply facility data in the annual reports (Hitachi city enterprise bureau, 2022; Hiroshima city waterworks bureau, 2022; Saitama city waterworks bureau, 2022; Chiba prefecture waterworks bureau, 2022) published by the water supply companies in the four representative areas, the average values of the water pumping capacity, rated output, pump head, number of pumps, and water volume in the distribution reservoir per water treatment plant in each representative area were

obtained and used as the pump specifications for the representative areas. Using these pump specifications, we created a hypothetical pump performance curve, and the power consumption characteristics of the pumps were determined as a linear function that varied with the number of pumps in operation, using the methods used in Arai et al. (2012), Hirano et al. (2017).

As shown “System Representation” No. 4 in Fig. 1, the number of the four types of water supply units was determined by dividing the water supply population per the water supply unit in each representative area by the water supply population in the four groups classified by clustering in each power system region. These methods are described in detail in Section 3.

2.3. Mathematical optimization models

In this study, three operational planning models that simulate the operation of the WDSs, generators in the grid, and interconnection lines among grids were formulated as MILPs and analyzed by optimization calculations. All the following models perform optimization calculation for 24 h with a one-hour resolution, and the evaluation for one year is performed by executing these one-day optimization 365 times. These MILP models were optimized using Gurobi Optimizer 9.5, a mathematical optimization solver.

2.3.1. Water pump model

Based on the WDS configuration determined in Section 2.2, pump operation is determined by minimizing the one-day power consumption of the pumps for each of the four types of water supply units in each system as the objective function. Using this model that minimizes power consumption, it is possible to calculate the power consumption of water pumps without DR operations.

2.3.2. Supply-demand analysis model

The model, developed with reference to Negishi et al. (2022), determines the operation of all generators in each power system and the interconnection lines among power systems while considering power flexibility through interconnection lines among power systems, with the objective function of minimizing the fuel cost for one day. Since WDS operation cannot be optimized with this model, the optimal operation of the power system is calculated by adding the power demand calculated by the Water Pump Model to other underlying power demands.

2.3.3. Supply-demand analysis and water pump model

This model combines the Supply-Demand Analysis Model with the Water Pump Model. The model determines the operation of various generators, interconnection lines, and water pumps for each system in Japan, using the minimization of the one-day fuel cost as the objective function. In this model, the constraints are that DR operation can only be performed within the controllable range of the pump, and also that the demand for tap water can be supplied with sufficient margin. Therefore, it will not have any negative impact on the water supply business or the power system reliability. By incorporating WDS operation into power system operation optimization, WDS operation hours can be determined in a way that minimizes fuel costs for the power system as a whole. Furthermore, when WDSs provide LFC regulating reserves, it is possible to reduce the amount of LFC regulating reserves that existing thermal power must provide, leading to more economically efficient operational results.

Table 1

Contribution to the system by water pumps in each case.

Case	Shift Operation timing	Provide LFC Balancing reserves
A	–	–
B	✓	–
C	✓	✓

2.4. Computational model analysis

To clarify the economic and environmental contributions of changing the operation method of water pumps, we divided the water pump operation methods into three cases: Cases A, B, and C. The calculation flow for each case is described as follows.

Case A In Case A, calculations were performed using the Water Pump Model and the Supply-Demand Analysis Model. First, assuming that the current water pumps in WDSs are being operated with the aim of saving energy, the Water Pump Model calculates the power consumption when the water pump operates at the lowest power consumption. Next, the calculated power consumption of the water pumps is added to other power demand data, and using this as input data, calculations are performed using the power system Supply-Demand Analysis Model. In this case, the efficiency of the power system operations without water pump DR operations was calculated. In this study, the DR effects were analyzed using this as a comparison target.

Case B The calculation model used is the Supply-Demand Analysis and Water Pump Model, which incorporates the Water Pump Model into the Supply-Demand Analysis Model. In this model, the operation of the water pumps is determined by minimizing the fuel cost of the entire power system. If there is a surplus in PV generation during daytime in sunny days, operating pumps during the day allows for more effective use of VRE and reduces fuel costs required for thermal power generators. Therefore, the water pump operations can be expected to shift to reduce the output curtailment of VRE generation. By comparing Case B with Case A, it is possible to clarify the DR effects of shifting the demand time using the pump operation in WDSs.

Case C In Case C, regarding the constraints on providing LFC regulating reserves in the same model as Case B, we assume that the LFC regulating reserves can be provided by inverter control from the water pump in WDSs without changing the number of pumps in operation. By taking over the LFC regulating reserves that used to be provided by thermal power generators, it is possible to reduce the operation of thermal power generators and contribute to reductions in fuel costs and CO₂ emissions. By comparing Case C with Case B, it is possible to clarify the DR effects when adding a function to provide LFC regulating reserve to the inverter of the water pumps in WDSs.

Table 1 summarizes the calculation conditions for each case. Differences in model settings depending on calculation conditions are described in Section 4, and calculation results comparing cases are described in Section 5.

3. Representation of WDS in Japan

3.1. Determination of representative areas

Fig. 3 shows the elevation difference and population density of the water supply areas of 1335 water service providers nationwide, obtained using the GIS from the Ministry of Land, Infrastructure, Transport, and Tourism’s data on water supply facilities (Ministry of Land,

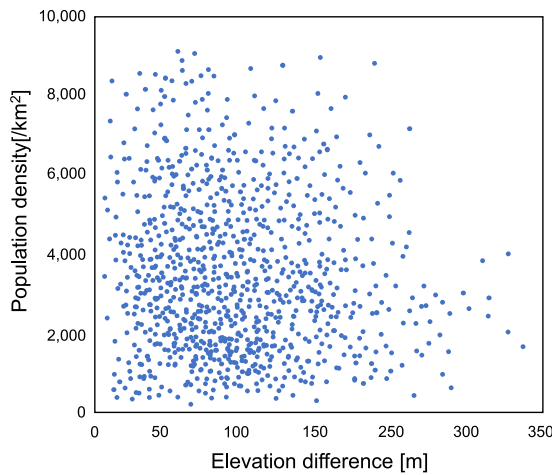


Fig. 3. Elevation difference and population density of each water supply area.

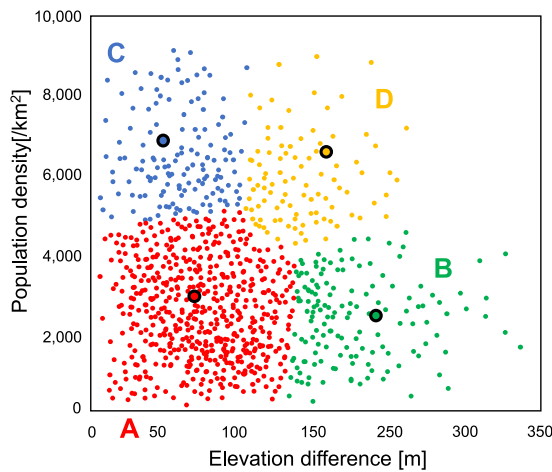


Fig. 4. Elevation difference and population density of each water supply area (clustered).

Infrastructure, Transport and Tourism, 2012) and the 5th order mesh data on elevation and slope (Ministry of Land, Infrastructure, Transport and Tourism, 2011). As shown in Fig. 4, the four points highlighted are the points closest to the last center of gravity in the clustering, and these four points are the representative areas of groups A to D, as follows.

- A: Hitachi city enterprise bureau
- B: Hiroshima city waterworks bureau
- C: Saitama city waterworks bureau
- D: Chiba prefecture waterworks bureau

Table 2 shows the average value of water pumps, rated output, volume of water delivered, pump head, and volume of water in distribution reservoirs per water treatment plant in each area, and the assumed population of water supply obtained from the annual business reports (Hitachi city enterprise bureau, 2022; Hiroshima city waterworks bureau, 2022; Saitama city waterworks bureau, 2022; Chiba prefecture waterworks bureau, 2022) of these water suppliers. The assumed water supply population is the water supply population per water treatment plant in each area. Next, by dividing the water supply population in the regional power system that corresponds to the four units clustered in Fig. 4 by the assumed water supply population by unit in Table 2 for each power system, the number of water supply units for each regional power system in Japan was obtained (Table 3).

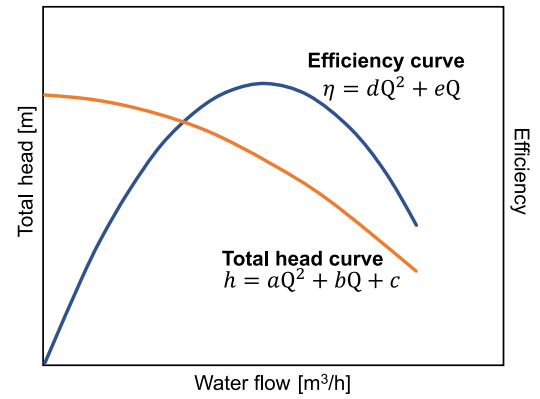


Fig. 5. Assumed characteristic curve of water pump.

3.2. Power consumption characteristic model of water pumps

We considered the power consumption characteristics of these four groups of water pumps. In this study (Fig. 5), we assume that the total head h [m] and efficiency η of the water pumps are expressed as quadratic curves for the amount of water delivered Q , as shown in Eq. (1) and (2), respectively:

$$h = aQ^2 + bQ + c \quad (1)$$

$$\eta = dQ^2 + eQ \quad (2)$$

When $Q = 0$, h has the maximum value $c = 1.2h_R$, and the point is the vertex of the quadratic curve; when $Q = Q_R$, η has a maximum value of 0.8. In this case, the values of the constants a-e for groups A to D can be obtained from Table 2 and Eq. (1) and (2) as presented in Table 4.

Next, we consider the power consumption characteristics when an arbitrary number of water pumps are connected in parallel to the water pumps. The methods for formulating this in MILP are described in Refs. Arai et al. (2012) and Hirano et al. (2017). First, the head h_n when the water pumps are operated and the pipeline characteristics from the water pumps to the distribution reservoir are approximated by a quadratic function as shown in Eq. (3) and (4) respectively:

$$h_n = a \left(\frac{Q}{n} \right)^2 + b \left(\frac{Q}{n} \right) + c \quad (3)$$

$$h = rQ^2 + H \quad (4)$$

where r [h^2/m^5] is the pipe resistance coefficient and H [m] is the actual head of the water pump. If these equations are solved for Q , the amount of water delivered Q_n when n pumps are operated can be obtained as shown in Eq. (5).

$$Q_n = \frac{b + \sqrt{b^2 - 4(n^2 r - a)(H - c)}}{2(n^2 r - a)} n \quad (5)$$

From the work of pumping water, if the power consumption of the pumps when n water pumps are operated is P_n [kW], the density of water is ρ [t/m^3] and the gravitational acceleration g [m/s^2].

$$P_n = \frac{1}{3600} \frac{\rho g h_n Q}{\eta_n} \quad (6)$$

Moreover, the efficiency η_n when water pumps are operated is

$$\eta_n = d \left(\frac{Q}{n} \right)^2 + e \left(\frac{Q}{n} \right) \quad (7)$$

From Eqs. (4), (6), and (7), the power consumption of each pump pattern can be obtained when the amount of water delivered Q changes.

Next, the power consumption characteristics are linearly regressed in the form $P = \alpha Q + \beta$. In Fig. 6, the pump pattern is determined

Table 2
Specifications of water supply unit based on the annual business report of each area.

Unit	Pump parameter		No. of pump	Reservoir capacity V [m ³]	Assumed water supply population
	Water flow P_R [m ³ /h]	Head h_R [m]			
A	955	39.6	3	20,349	61,521
B	836	58.1	3	8,476	60,529
C	1385	35.2	5	30,465	90,262
D	1524	49.5	4	47,451	180,239

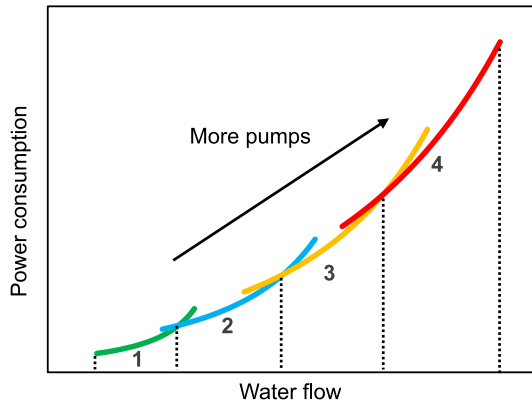


Fig. 6. Operating properties of multiple pumps.

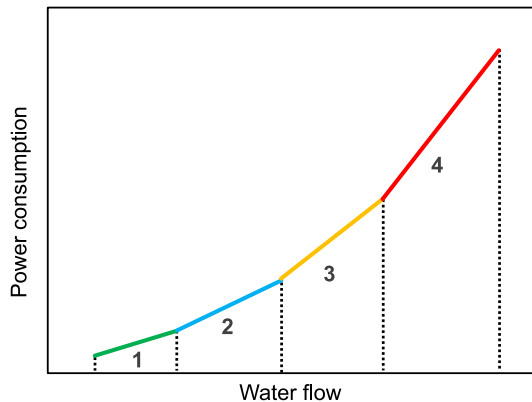


Fig. 7. Operating properties of multiple pumps (Linear regressed).

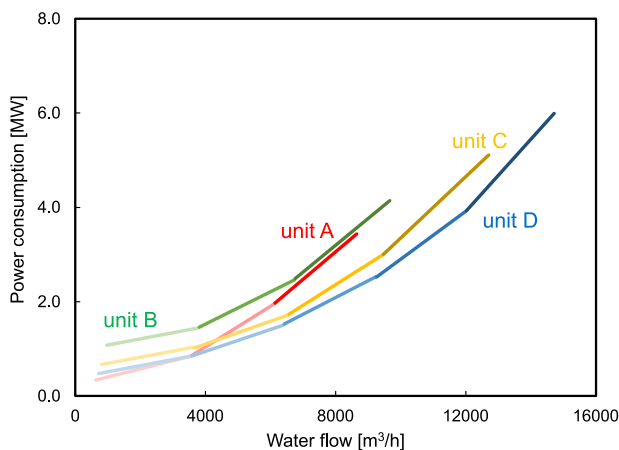


Fig. 8. Approximated operation properties of water pumps in each water supply unit.

Table 3

Assumed Number of each water supply unit for every regional power system in Japan.

	Unit A	Unit B	Unit C	Unit D
Hokkaido	6	55	1	2
Tohoku	68	162	13	6
Tokyo	54	69	61	43
Chubu	50	55	18	5
Hokuriku	41	36	10	3
Kansai	38	63	21	6
Chugoku	24	23	6	4
Shikoku	11	22	3	1
Kyushu	38	62	14	3
Okinawa	10	0	0	0

Table 4

Calculated value of constants a-e.

Unit	a $\times 10^{-8}$ [m ⁻⁵ h ²]	b [m ⁻² h]	c [m]	d $\times 10^{-8}$ [m ⁻⁶ h ²]	e $\times 10^{-5}$ [m ⁻³ h]
A	-2.46	0	47.5	-2.74	9.34
B	-2.35	0	69.7	-3.26	8.44
C	0.95	0	42.2	-1.55	3.75
D	4.02	0	59.4	4.76	0.92

such that the power consumption is the lowest in relation to the amount of water delivered. The lower and upper limits of the amount of water delivered in each pump pattern (i.e., the amount of water delivered at the intersection of the power consumption characteristics of adjacent pump patterns) and linearly regress the power consumption characteristics for each pump pattern in that section (Fig. 7). When the power consumption characteristics of the pumps in units A to D are obtained as described, the power consumption characteristics are as shown in Fig. 8. In the calculations, water density is $\rho = 1.0$ [t/m³], gravitational acceleration is $g = 9.8$ [m/s²], and the pipe resistance coefficient is $r = 1.0 \times 10^{-7}$ [h²/m⁵].

4. Mathematical models and used data

4.1. Water pump model

This model is a calculation model developed with reference to Negishi and Ikegami (2019) and simulates the case where water pumps are operated in such that they minimize their power consumption in the form of an MILP. As described in Section 2.3.1, this model optimizes the number of water pumps in operation and the amount of water delivered at each time by considering the capacity of the distribution reservoirs, the demand for water distribution, and the power consumption of the water pumps when they are in operation for the operation of 10 water pump systems throughout Japan modeled as water supply units. The model was run every 12 h for one year with a one-hour time resolution to minimize the power consumption of the water pumps. As with the Supply-Demand Analysis Model, optimization is performed for 24 h to consider the next day's water volume in the distribution reservoir, and the data from the 12th hour in that calculation is used as the initial values for the next 24-h calculation.

The objective function and constraints are provided by Eq. (8)–(14). The nomenclature is presented in Table 5.

Table 5
Nomenclature used in the Water Pump Model.

[Constants]	
$gNum$	Total number of area
$hNum$	Total number of time-slot in a day [h]
$wtNum$	Total number of types of water supply unit
$wpNum$	Total number of pump patterns in the selected water supply unit
$wuNum$	Total number of water supply unit
α	Linear regression coefficient for power consumption characteristics of water pump
β	Linear regression coefficient for power consumption characteristics of water pump
qLB	Lower bound of volume of the reservoir [m ³]
qUB	Upper bound of volume of the reservoir [m ³]
qdm	Water supply demand [m ³]
$iQwf$	Initial quantity of water flow from a single water supply unit in a day [m ³ /h]
$iWdst$	Water supply demand [m ³]
[Index]	
g	Area
h	Time-slot in a day [h]
wt	Water supply unit type (unit A = 0, unit B = 1, unit C = 2, unit D = 3)
wp	Pump pattern
[Decision variables]	
$PSUM$	Power consumption in each type of water supply unit [MW]
PWC	Power consumption of water pump in each pump pattern [MW]
PAT	Indication variable of pump pattern (binary)
QWF	Water flow from a single water supply unit [m ³ /h]
$W DST$	Initial Amount of water in reservoir of a single water supply unit in a day [m ³]

(Objective function)

$$\min. \sum_{g=0}^{gNum} \sum_{h=1}^{hNum} \sum_{wt=0}^{wtNum} (PSUM_{g,h,wt}) \quad (8)$$

(Constraints)

$$\sum_{wp=1}^{wpNum_{wt}} PAT_{g,h,wt,wp} \leq 1 \quad (9)$$

$$PSUM_{g,h,wt} = \sum_{wp=1}^{wpNum_{wt}} PWC_{g,h,wt,wp} \quad (10)$$

$$PWC_{g,h,wt,wp} = (\alpha_{wt,wp} \cdot QWF_{g,h,wt,wp} + \beta_{wt,wp} \cdot PAT_{g,h,wt,wp}) \cdot 0.001 \cdot wuNum_{g,wt} \quad (11)$$

$$qLB_{wt,wp} \cdot PAT_{g,h,wt,wp} \leq QWF_{g,h,wt,wp} \leq qUB_{wt,wp} \cdot PAT_{g,h,wt,wp} \quad (12)$$

$$W DST_{g,h,wt} = W DST_{g,h-1,wt} + 1 \cdot \left(\sum_{wp=1}^{wpNum_{wt}} QWF_{g,h,wt,wp} - qdm_{h,wt} \right) \quad (13)$$

$$\sum_{wp=1}^{wpNum_{wt}} QWF_{g,0,wt,wp} = iQwf_{g,wt} \quad (14)$$

$$W DST_{g,0,wt} = iW dst_{g,wt} \quad (15)$$

Eq. (8) is the objective function, which represents the power consumption of the water pumps in all grids over 24 h. Eq. (9) represents the operating pattern constraint of the water pump. Any pump pattern can be selected at any time, including no pump patterns. Eq. (10) represents the amount of power consumed by the water pump in each water supply unit. Eq. (11) is an equation constraint for the power consumption characteristics of the pump pattern. Eq. (12) represents the upper and lower limits of the water pump capacity. Eq. (13) represents a constraint on the time variation in the water volume in the distribution reservoir. The water volume in the distribution reservoir was calculated by subtracting the volume of water demand from the volume of water in the reservoir at the previous time and adding the volume of water delivered from the water pump. Eq. (14) and (15)

represent a transfer constraint on the amount of water delivered and the amount of water in the reservoir 12 h before the final time, respectively.

4.2. Supply-demand analysis model

As mentioned in Section 2.3.2, the model was developed according to Negishi et al. (2022). The model considers the interconnection lines among power grids and optimizes the operation of generators such as thermal power plants and the interconnection lines for 10 power grids throughout Japan every 12 h for one year with a time resolution of 1 h to minimize fuel costs. In addition to the Water Pump Model, in the one-year optimization calculation, a 24-h optimization is performed to consider the next day's operation of pumped-storage power generation, and the data from the 12th hour of the calculation are used as the initial value for the next 24-h calculation.

In this section, we describe the objective function and the main constraints. The nomenclature is presented in Table 6. The aforementioned symbols are not listed in Table 6.

(Objective function)

$$\min. \sum_{g=0}^{gNum} \sum_{u=1}^{tpNum_g} \sum_{h=1}^{hNum} (GTP_{g,u,h} \cdot fcsA_{g,u} + UTP_{g,u,h} \cdot fcsB_{g,u} + UON_{g,u,h} \cdot fcsS_{g,u}) \quad (16)$$

(Constraints)

$$\begin{aligned} dmd_{g,h} &= \sum_{u=1}^{tpNum_g} (1 - pAux_{g,u}) \cdot GTP_{g,u,h} \\ &+ \sum_{u=1}^{stNum_g} (PMG_{g,u,h} - PMS_{g,u,h}) \\ &- CPV_{g,h} - CWP_{g,h} \\ &+ \sum_{n=1}^{nNum} (GIC_{n,h} \cdot (1 - nLss_n) ito_{n,g} - GIC_{n,h} \cdot ifr_{n,h}) \\ &+ gpv_{g,h} + gwp_{g,h} + ghedr_{g,h} \end{aligned} \quad (17)$$

$$dmlup_{g,h} = dmd_{g,h} \cdot dmlfc_{UP_{g,m}} \quad (18)$$

$$dmlup_{g,h} \leq \sum_{u=1}^{tpNum_g} (1 - pAux_{g,u}) LTP_{UP_{g,u,h}}$$

Table 6
Nomenclature used in the Supply-Demand Analysis Model.

[Constants]	
$tpNum$	Total number of thermal power unit
$stNum$	Total number of generators in pumped-storage unit
$nLss$	Energy loss rate of interconnection lines
ifr	Indication variable of interconnection transmitted from(binary)
ito	Indication variable of interconnection transmit to(binary)
$fcsA$	Coefficient of fuel cost function [JPY/MW]
$fcsB$	Coefficient of fuel cost function [JPY]
$fcsS$	Start-up cost of each unit [JPY]
$pAux$	Auxiliary power ratio
$dmnd$	Load [MW]
gpv	Output of PV without curtailment [MW]
gwp	Output of WP without curtailment [MW]
$ghydr$	Output of hydro generator [MW]
$dmlup$	Required upward balancing reserve of demand [MW]
$pulup$	Required upward balancing reserve of PV [MW]
$wplup$	Required upward balancing reserve of WP [MW]
$hydr_{lfc}$	Balancing reserve capacity provided by hydrogen generators [MW]
$dmLfc_{UP}$	Upward fluctuation rate of demand
$pvLfc_{DW}$	Downward fluctuation rate of PV
$wplfc_{DW}$	Downward fluctuation rate of WP
[Index]	
u	Generation unit including clustered generation units
[Decision variables]	
GTP	Output of thermal generator [MW]
UTP	State of thermal generator (integer variables)
UON	Start-up flag of thermal generator (integer variables)
PMG	Output of pumped-storage generator [MW]
PMS	Load of pumped-storage generator [MW]
CPV	Curtailed output of PV [MW]
CWP	Curtailed output of WP [MW]
GIC	Power flow of interconnection line [MW]
LTP_{UP}	Upward balancing reserve capacity provided by thermal generators [MW]
LP_{GP}	Upward balancing reserve capacity provided by pumped storage generators during generating operation [MW]
LIC_{DW}	Downward power flow fluctuation which is occurred by exchanging balancing reserves [MW]
$PLIC_{UP}$	Upward balancing reserve provided to an area linked by interconnection-line [MW]
$PLIC_{DW}$	Downward balancing reserve provided to an area linked by interconnection-line [MW]

$$\begin{aligned}
& + \sum_{u=1}^{stNum_g} (LPG_{UP_{g,u,h}} + LTP_{UP_{g,u,h}}) \\
& + \sum_{n=1}^{nNum} ((ifr_{n,g} + ito_{g,n}(1 - nLss_n)) \\
& \cdot (LIC_{DW_{n,h}} - PLIC_{UP_{g,n,h}} - PLIC_{DW_{g,n,h}})) \\
& + hydr_{lfc_{g,h}}
\end{aligned} \quad (19)$$

$$\begin{aligned}
& + \sum_{u=1}^{stNum_g} (LPG_{UP_{g,u,h}} + LTP_{UP_{g,u,h}}) \\
& + \sum_{n=1}^{nNum} ((ifr_{n,g} + ito_{g,n}(1 - nLss_n)) \\
& \cdot (LIC_{DW_{n,h}} - PLIC_{UP_{g,n,h}} - PLIC_{DW_{g,n,h}})) \\
& + ghydr_{lfc_{g,h}}
\end{aligned} \quad (23)$$

$$pv_{lup_{g,h}} = (gpv_{g,t} - CPV_{g,m})pvLfc_{DW_{g,m}} \quad (20)$$

$$\begin{aligned}
pv_{lup_{g,h}} & \leq \sum_{u=1}^{tpNum_g} (1 - pAux_{g,u}) LTP_{UP_{g,u,h}} \\
& + \sum_{u=1}^{stNum_g} (LPG_{UP_{g,u,h}} + LTP_{UP_{g,u,h}}) \\
& + \sum_{n=1}^{nNum} ((ifr_{n,g} + ito_{g,n}(1 - nLss_n)) \\
& \cdot (LIC_{DW_{n,h}} - PLIC_{UP_{g,n,h}} - PLIC_{DW_{g,n,h}})) \\
& + hydr_{lfc_{g,h}}
\end{aligned} \quad (21)$$

$$wpl_{up_{g,h}} = (wptv_{g,t} - CWP_{g,m})wplfc_{DW_{g,m}} \quad (22)$$

$$wpl_{up_{g,h}} \leq \sum_{u=1}^{tpNum_g} (1 - pAux_{g,u}) LTP_{UP_{g,u,h}}$$

The objective function in Eq. (16) represents the total fuel and startup costs of nuclear and thermal generators over 24 h. Eq. (17) matches the hourly power demand and supply in each system. The constraint is that the amount of electricity generated by nuclear power, thermal power, PV power, WP, hydroelectric power, pumped-storage power, and the amount of electricity received by the interconnection line minus the amount of output curtailment of PV power, WP, power consumption of pumped-storage pumps, and transmission by the interconnection line equals the amount of electricity demand. Eqs. (18)–(22) are the constraints on the upward balancing reserves of each generator. Eqs. (18), (20), and (22) represent the required upward balancing reserves for electricity demand, PV power generation, and WP generation, respectively. Eqs. (19), (21), and (21) are the constraints to ensure the required upward balancing reserves for each fluctuation caused by the electricity demand and output of PV and WP. The required balancing reserves for electricity demand is assumed to be 2% of electricity demand (Negishi et al., 2022); for PV power generation, 10% of the output generation (Negishi et al., 2022); and for WP generation, it varies in percentage depending on the amount of WP generation (Ikegami et al., 2018).

Similar constraints are imposed on the balancing reserves caused by the electricity demand and output of the PV and WP.

Table 7
Nomenclature used in Water Pump Model.

[Constants]	
$pload$	Minimized power consumption of water pumps in entire area calculated in the Water Pump Model [MW]
[Decision variables]	
LPM_{UP}	Upward balancing reserve capacity provided by water pumps [MW]
LPM_{DW}	Downward balancing reserve capacity provided by water pumps [MW]
Q_{UP}	Difference in the water flow when a single water supply unit provides upward balancing reserve [m^3/h]
Q_{DW}	Difference in the water flow when a single water supply unit provides downward balancing reserve [m^3/h]
$WSTO_D$	Difference in the quantity of water of reservoir when a single water supply unit provides balancing reserve [m^3]

4.3. Supply-demand analysis and water pump model

The Supply-Demand Analysis and Water Pump Model is a calculation model that incorporates the Water Pump Model into the Supply-Demand Analysis Model and further improves it so that the water pumps can provide balancing reserves to the power system. In this model, the constraints of the Water Pump Model are added to the Supply-Demand Analysis Model, and the operation of the water pump is optimized simultaneously with the optimization of the operation of each generator and the electricity supply through the interconnection. Because the objective function of the model is to minimize fuel costs, as in the electricity supply and demand analysis model, the water pumps are expected to be operated to minimize the fuel costs of the entire power system, that is, to reduce the output curtailment of PV and WP output and to provide regulating reserves to the power system.

The objective function is the same as that of the Supply-Demand Analysis Model, as shown in Eq. (24).

(Objective function)

$$\min. \sum_{g=0}^{gNum} \sum_{u=1}^{tpNum_g} \sum_{h=1}^{hNum} (GTP_{g,u,h} \cdot fcsA_{g,u} + UTP_{g,u,h} \cdot fcsB_{g,u} + UON_{g,u,h} \cdot fcsC_{g,u}) \quad (24)$$

The constraints are the conditions of the Supply-Demand Analysis Model with the addition of the conditions shown in Section 4.1. The following is a list of the changed and newly added constraint conditions. The nomenclature is listed in Table 7. The aforementioned symbols are not listed in Table 7.

$$\begin{aligned} dmnd_{g,h} = & \sum_{u=1}^{tpNum_g} (1 - pAux_{g,u}) \cdot GTP_{g,u,h} \\ & + \sum_{u=1}^{stNum_g} (PMG_{g,u,h} - PMS_{g,u,h}) \\ & - CPV_{g,h} - CWP_{g,h} \\ & + \sum_{n=1}^{nNum} (GIC_{n,h} \cdot (1 - nLss_n) ito_{n,g} - GIC_{n,h} \cdot ifr_{n,h}) \\ & + gpv_{g,h} + gwp_{g,h} + ghydr_{g,h} \\ & + pload_{g,h} - \sum_{wt=0}^{wtNum} \sum_{wp=0}^{wpNum_{wt}} (PWC_{g,h,wt,wp}) \end{aligned} \quad (25)$$

$$\begin{aligned} dmlup_{g,h} = & \left(dmnd_{g,h} - pload_{g,h} + \sum_{wt=0}^{wtNum} \sum_{wp=0}^{wpNum_{wt}} (PWC_{g,h,wt,wp}) \right) \\ & \cdot dMLfc_{UP_{g,m}} \end{aligned} \quad (26)$$

$$dmlup_{g,h} \leq \sum_{u=1}^{tpNum_g} (1 - pAux_{g,u}) LTP_{UP_{g,u,h}}$$

$$\begin{aligned} & + \sum_{u=1}^{stNum_g} (LPG_{UP_{g,u,h}} + LTP_{UP_{g,u,h}}) \\ & + \sum_{n=1}^{nNum} (ifr_{n,g} + ito_{g,n}(1 - nLss_n)) \\ & \cdot (LIC_{DW_{n,h}} - PLIC_{UP_{g,n,h}} - PLIC_{DW_{g,n,h}}) \\ & + hydr_{fc_{g,h}} \\ & + \sum_{wt=1}^{wtNum} \sum_{wp=1}^{wpNum_{wt}} LPM_{UP_{g,h,wp}} \end{aligned} \quad (27)$$

$$Q_{UP_{g,h}} = \sum_{wt=1}^{wtNum} \sum_{wp=1}^{wpNum_{wt}} \frac{LPM_{UP_{g,h,wp}}}{\alpha_{wt,wp}} \cdot 1000 \quad (28)$$

$$Q_{DW_{g,h}} = \sum_{wt=1}^{wtNum} \sum_{wp=1}^{wpNum_{wt}} \frac{LPM_{DW_{g,h,wp}}}{\alpha_{wt,wp}} \cdot 1000 \quad (29)$$

$$WDST_{D_{g,h}} = \sum_{t=1}^h (Q_{UP_{g,t,wt}} - Q_{DW_{g,t,wt}}) \quad (30)$$

$$sLB_{wt} \leq WDST_{g,h,wt} + WDST_{D_{g,h-1,wt}} - Q_{DW_{g,h,wt}} \quad (31)$$

$$sLB_{wt} WDST_{g,h,wt} - Q_{DW_{g,h,wt}} \quad (32)$$

$$sUB_{wt} \geq WDST_{g,h,wt} + WDST_{D_{g,h-1,wt}} + Q_{UP_{g,h,wt}} \quad (33)$$

$$sUB_{wt} \geq WDST_{g,h,wt} + Q_{UP_{g,h,wt}} \quad (34)$$

$$\begin{aligned} LPM_{DW_{g,h,wt,wp}} \leq & PWC_{g,h,wt,wp} \\ & - (\alpha_{wt,wp} \cdot qLB_{wt,wp} + \beta_{wt,wp}) PAT_{g,h,wt,wp} \\ & \cdot 0.001 \cdot wuNum_{g,wt} \end{aligned} \quad (35)$$

$$\begin{aligned} LPM_{DW_{g,h,wt,wp}} \leq & \left(\alpha_{wt,wp} \cdot \frac{qUB_{wt,wp} + qLB_{wt,wp}}{2} + \beta_{wt,wp} \right) \\ & \cdot PAT_{g,h,wt,wp} \cdot 0.2 \end{aligned} \quad (36)$$

$$\begin{aligned} LPM_{UP_{g,h,wt,wp}} \leq & (\alpha_{wt,wp} \cdot qUB_{wt,wp} + \beta_{wt,wp}) \\ & \cdot PAT_{g,h,wt,wp} \cdot 0.001 \cdot wuNum_{g,wt} \\ & - PWC_{g,h,wt,wp} \end{aligned} \quad (37)$$

$$\begin{aligned} LPM_{UP_{g,h,wt,wp}} \leq & \left(\alpha_{wt,wp} \cdot \frac{qUB_{wt,wp} + qLB_{wt,wp}}{2} + \beta_{wt,wp} \right) \\ & \cdot PAT_{g,h,wt,wp} \cdot 0.2 \end{aligned} \quad (38)$$

Eq. (25) matches the hourly power demand and supply in each system. To optimize the operation of the water pumps in the model, we added the power consumption of the water pumps, determined in the model, to Eq. (17) by subtracting the power consumption of the water pumps when operated under the power consumption minimization obtained by the Water Pump Model from the power demand of the entire grid. Eqs. (26) and Eq. (27) are the constraints on upward

Table 8
Installed capacity of VRE units in Japan for 2040 in each scenario.

	Scenario 1		Scenario 2	
	PV [MW]	WP [MW]	PV [MW]	WP [MW]
Hokkaido	4,320	2,603	8,640	5,206
Tohoku	13,736	12,180	27,472	24,360
Tokyo	46,899	9,257	93,798	18,514
Chubu	30,170	5,651	60,340	11,302
Hokuriku	3,065	749	6,130	1,498
Kansai	21,130	2,145	42,260	4,290
Chugoku	15,394	1,998	30,788	3,996
Shikoku	8,138	1,540	16,276	3,080
Kyushu	30,766	2,965	61,532	5,930
Okinawa	1,587	312	3,174	624

balancing reserves. As with Eq. (25), $pload$ is subtracted from $dmnd$, and PWC is added. In addition, the balancing reserves provided by the water pumps, LPM_{UP} and LPM_{DW} , are added to the term of provided balancing reserves. Eqs. (28) and (29) represent the constraints on the changes in the amount of water delivered as the water pump provides balancing reserves. Eq. (30) represents the change in reservoir volume when the pumps provide balancing reserves. Eqs. (31)–(34) are constraints on the upper and lower limits of the water reservoirs. Eqs. (32) and (34) are constraint equations when considering the case in which the balancing reserves have been provided by the previous time slot. Eqs. (35)–(38) indicate the constraints on the upper limits of the upward and downward balancing reserves that can be provided by water pumps. The value of the upper limit is set to the extent that the amount of water delivered by the water pump after providing balancing reserves does not exceed the amount of water delivered in the pump pattern currently in operation and to 20% of the median power consumption in the pump pattern at that time.

4.4. Calculation conditions

Calculation conditions in the analyses are explained as follows. This study analyzes 10 regions of Japan using a case study assuming the year 2040, when the introduction of VRE sources in Japan is expected to progress and output curtailment of VRE generation will become more pronounced than it is today. For all cases, two scenarios were prepared with different installed VRE source capacities.

Scenario 1 This scenario of penetration of VRE generators is based on the target values in the medium case (Ministry of the Environment, 2015). The target values of RE penetration in 2040 are set in terms of total nationwide installed capacity, and these values must be allocated among the regions to obtain input data for the proposed model. Regarding the allocation method, shares of each region in the total installed capacity were calculated based on the data as of the end of 2016 for VRE sources other than WP (Agency for natural Resources and Energy, 2022) and with reference to Japan Wind Power Association data for WP sources (Japan Wind Power Association, 2014).

Scenario 2 In this scenario, the installed capacity value is twice that in Scenario 1 to examine the operation of the water pumps and the resulting effects when additional renewable energy is installed.

The installed capacity of the VRE for every regional power system for 2040 is listed in Table 8. The optimization model for each case was calculated using Gurobi Optimizer 9.5, a mathematical optimization solver. In all cases, the parameter for the deviation of the cost function value from the linear relaxation problem of integer programming was set to 0.5%. As a result, in our model, there are a total of 72,048 decision variables, of which 27,744 are integer variables. Additionally, the model includes 84,776 constraints.

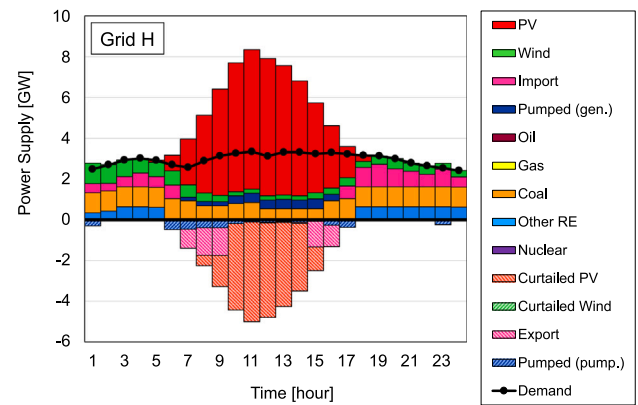


Fig. 9. Results of generators operation (Scenario 1, Shikoku area, May 20, 2040).

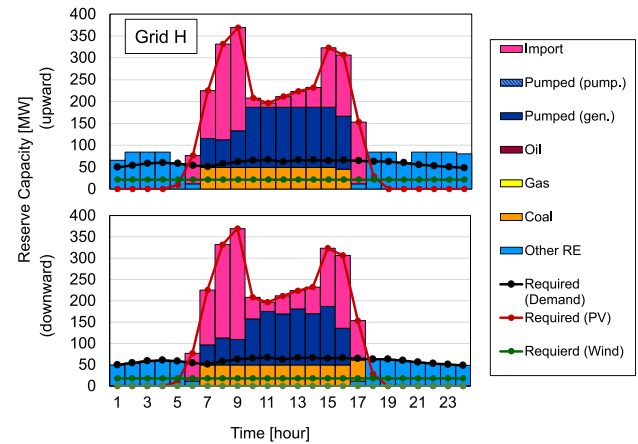


Fig. 10. Results of required and supplied balancing reserves (Scenario 1, Shikoku area, May 20, 2040).

5. Calculation results

5.1. Changes in pump operations

This section presents how the operation of water pumps changed over one day by comparing the results of the operation of water pumps on representative days in Cases A and C. As an example of the analysis results, we show those for Scenario 1 in the Shikoku area on May 20, 2040.

Figs. 9 and 10 show the results of the analysis of electricity supply and demand for Case A, operated under the minimization of water pump power consumption. First, Fig. 9 shows that PV generation increased from 6:00 on this day, and PV output was suppressed from 8:00 to 15:00 owing to the oversupply of electricity. Fig. 10 shows that to supply the required balancing reserves of PV, pumped-storage generators, coal-fired power generators, and imported balancing reserves are used from 7:00 to 16:00.

Fig. 11 shows the time variation in the total power consumption of all water pumps in the Shikoku area on that day. Time trends of the usage of water reservoirs are shown in Fig. 12. In Fig. 11, in Case C, power consumption is higher than in Case A from 8:00 to 12:00 and 14:00 to 15:00, when the power output is suppressed, and power consumption is lower during the other periods. This result indicates that the water pumps shift their operation to reduce the output curtailment during the period when output curtailment occurs.

Furthermore, in Case C, up/downward balancing reserves are provided by the pumps, and the operation is shifted to secure adjusting

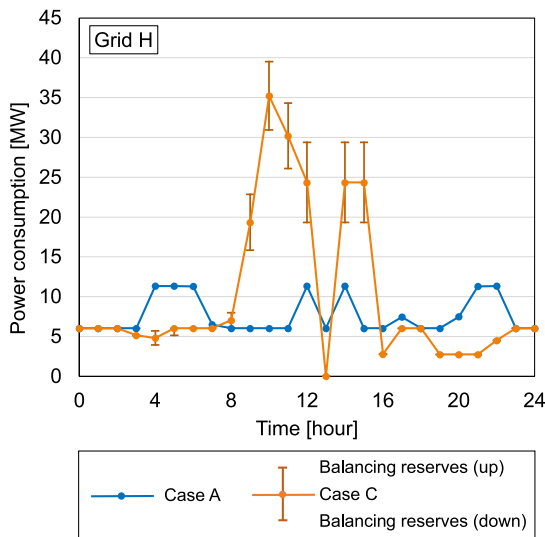


Fig. 11. Power consumption of water pumps at each time (Scenario 1, Shikoku area, May 20, 2040).

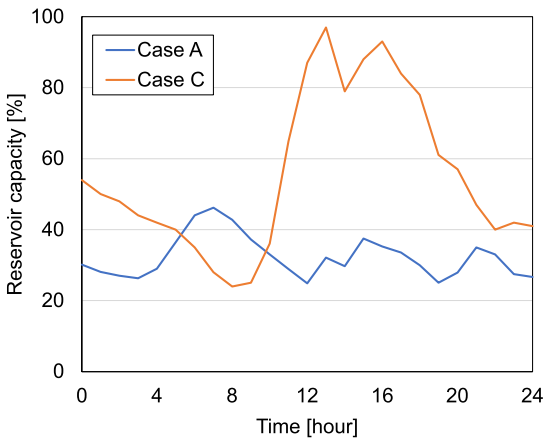


Fig. 12. Reservoir capacity of entire units at each time (Scenario 1, Shikoku area, May 20, 2040).

power in the up and down directions from 9:00 to 15:00, except at 13:00, when the required adjusting power due to the PV increases in the system.

Additionally, the power consumption of the pumps, which had remained above 15 MW since 9:00, decreased to 0 MW at 13:00, and then increased again to 24.5 MW at 14:00. This transition is thought to be observed because, as shown in Fig. 12, the water level in the reservoir increased to near the upper limit at 13:00, and the pumps were unable to pump any more water; thus, the power consumption decreased despite the occurrence of output curtailment, and then increased again at 14:00 when there was additional room in the reservoir capacity.

5.2. Supply–demand analysis results

This section presents the results of the VRE output curtailment, CO₂ emissions, and fuel costs calculated by analyzing the supply and demand of electricity throughout Japan over one year.

Tables 9, 10, 11 show the calculation results of annual VRE output curtailment, CO₂ emissions, and fuel costs for each scenario in Japan for each case and scenario. For Cases B and C, the percentage change compared with Case A is shown.

Table 9

Annual output curtailment of PV and WP in 2040 [GWh/year].

Scenario	Case A	Case B	Case C
1	22,982	22,553 (−1.4%)	22,101 (−2.8%)
2	35,952	34,981 (−1.9%)	34,117 (−3.2%)

Table 10

Annual CO₂ emissions in 2040 [Mton/year].

Scenario	Case A	Case B	Case C
1	395.1	390.6 (−1.1%)	390.0 (−1.3%)
2	273.0	269.8 (−1.2%)	268.4 (−1.7%)

Table 11

Annual fuel costs in 2040 [billion JPY/year].

Scenario	Case A	Case B	Case C
1	3918.5	3879.4 (−1.0%)	3855.8 (−1.6%)
2	2424.9	2390.9 (−1.4%)	2376.4 (−2.0%)

Table 12

Annual power consumption of water pumps in 2040 [GWh/year].

Scenario	Case A	Case B	Case C
1	2726.3	2748.1 (+0.8%)	2767.2 (+1.5%)
2	2726.3	2759.0 (+1.2%)	2786.3 (+2.2%)

First, in Case B, as mentioned in Section 5.1, the amount of VRE output curtailment was reduced by changing the operation timing of the water pumps and shifting the demand when the output curtailment of VRE occurred. In addition, the reduction rates in Scenario 2, in which the amount of installed renewable energy increased, were all higher than those in Scenario 1. This result occurs because the increase in the amount of renewable energy introduced increases the amount of output curtailment and the effect of shifting the operation timing of the water pumps. In addition, in Case C, the reduction in output curtailment is larger than that in Case B in both scenarios. This result occurs because water pumps provide balancing reserves to the grid, suppressing the operation of thermal power generators and promoting renewable energy use.

Table 12 shows the total power consumption of the water pumps for all water supply units in each scenario. In Case A, the pump power consumption was the same when the power consumption was the lowest, regardless of the scenario. We compared Cases B and C to Case A: the power consumption increases in both scenarios, and the rate of change is greater in Case C than in Case B. This result is thought to occur because in Case C, the change in operation with respect to Case A, where the pump power consumption was minimized, was greater than that in Case B.

In addition, as aforementioned, the annual output curtailment, CO₂ emissions, and fuel costs could be reduced for the entire system, despite the increase in pump power consumption, when comparing Case A with Cases B and C.

6. Conclusions

In this study, to clarify the potential of water pumps in the WDS to work with power systems in Japan, we first divided the current Japanese WDS into four groups based on two parameters: population density of the water supply population and elevation difference. Next, we used optimization calculation models to simulate two cases: one

case in which the water pumps are operated to minimize the power consumption of the pumps as the current operation method, and another case in which the operation timing of the pumps is changed to avoid suppressing the output of the VRE generation and to provide balancing reserves to the power system as necessary. In the optimization analysis, supply–demand analysis was conducted assuming the year 2040, when VRE generation is expected to be widely used, and the environmental and economic effects of the optimal operation of the water pumps for the power system were evaluated in terms of the amount of VRE output curtailment, CO₂ emissions, and annual fuel costs.

As a result, by optimizing the operation of the water pumps for the power system, the power consumption of the water pumps was higher than that of the current operation; however, the VRE output curtailment, CO₂ emissions, and annual fuel costs were reduced by 1% or more. These results show that an operation method that prioritizes the use of renewable energy, rather than an operation that minimizes power consumption, results in more economical and environmentally friendly operation. Furthermore, by controlling the power demand of water pumps, which account for less than 1% of Japan's total electricity demand, the ability to reduce generation fuel costs and CO₂ emissions by more than 1% has a significant impact. In addition, providing the system with balancing reserves from the water pumps resulted in an additional effect. It is meaningful from an economic and environmental perspective to proactively introduce DR operations that utilize water pumps at WDSs.

However, from the operation of the pumps obtained through the analysis, we observed that the capacity of the water reservoir connected to the pumps reached its maximum capacity when the output curtailment of VRE occurred, and the power consumption of the pumps was reduced during that time. This suggests that further benefits can be obtained by replacing water facilities with a larger water reservoir capacity, if the location allows. This point requires further analysis.

CRedit authorship contribution statement

Yushi Akimoto: Writing – original draft, Visualization, Resources, Data curation. **Shintaro Negishi:** Writing – review & editing, Project administration, Methodology, Conceptualization. **Takashi Ikegami:** Supervision, Software, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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